أسئلة سنوات سابقة

دوائر كهربائية ١

ابذل غاية الجهد ييسّر لك غاية الهدى
(والذين جاهدوا فينا لنهدينّهم سبلنا)



عضو الجمعية الطلابية:
محمد جاموس

الجمعية العلمية الطلابية

SPARK TEAM

Spark
POLYTECHNIC

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Problem 1.[10 points]

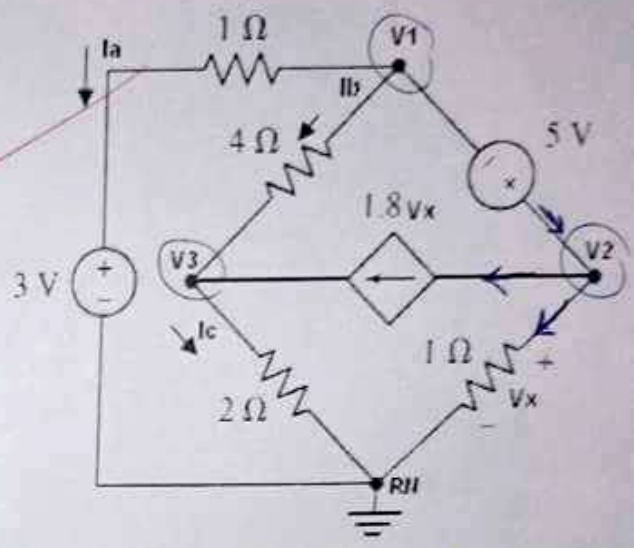
In the circuit of figure. Use nodal analysis method to find the nodal voltages and branch currents:

1- Write the required nodal equations (don't simplify)

(1) $\frac{V_1 - V_3}{4} + 1.8V_x = \frac{V_3}{2}$

(2) $I_a + I_a = \frac{V_2 - 0}{1}$

(3) $1.8V_x + \frac{V_1 - V_3}{4} = \frac{V_2}{1}$



2- Write the expressions for the branch currents:

Ia= $\frac{V_1 - 3}{1}$ = ~~0.7592 V~~ 3 = −4.647 − 3 = −1.647A

Ib= $\frac{V_1 - V_3}{4}$ = −0.98575

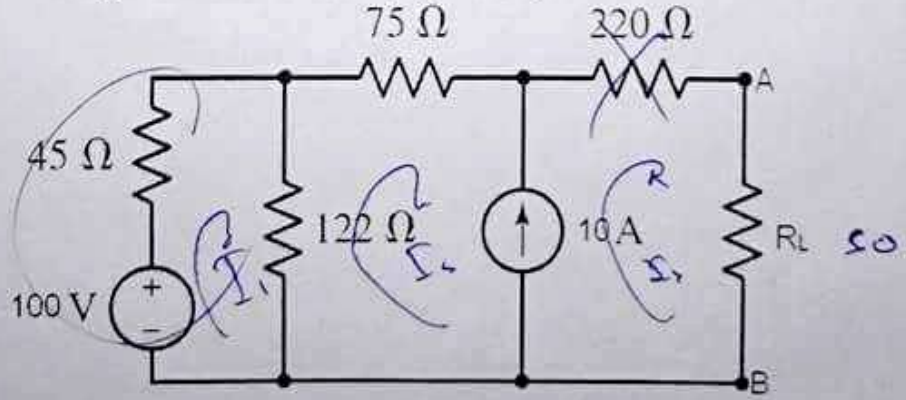
Ic= $\frac{V_3 - 0}{2}$ = $\frac{-0.704}{2}$ = −0.3522

Problem 2.[10 points]

For the circuit shown in figure find:

1- The Thevenin equivalent circuit with respect to terminals A-B.

2- The current through load resistance (R_L=50 Ω).



Good Luck

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Al Balqa' Applied University
Faculty of Engineering Technology
Electrical Engineering Department

كلية الهندسة التكنولوجية

Student Name: [redacted]

First Semester: 2013/2014
First Exam

#spark_team

3171502006 2

Course Name: Electrical Circuits (1)	Date: 2/11/2013
Instructors: Dr. Dwairi, Dr Abu Harb, Eng. Juma Abu-Ghalune	Exam Time: 50 Minutes

Problem 1: Consider the circuit shown where $i_s = 1A$.
Write the equations that are used to solve the problem



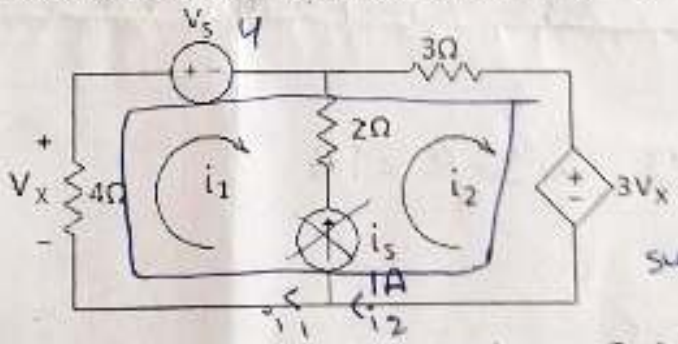
(2115) → 4Ω
sub(1,3) → 4Ω
$i_4 = \frac{1*4}{8} = \frac{1}{2}$

a Using current divider rule, find i_1 and v_2

$i_1 = .5 * \frac{20}{25} = .4A$, $v_2 = 20 * .1 = 2V$

b- Using voltage divider rule find v_1, $v_1 = \frac{2*3}{4+2} = \frac{3}{2}V$

Problem 2: Consider the circuit shown where $i_s = 1A$, and $v_s = 4\ V$.



supermesh

a- Using mesh analysis, write all the required equations to find i_1, i_2 and v_x

supermesh

1- $4i_1 + 3i_2 + 4 + 3v_x = 0 \rightarrow (1)$

$-8i_1 + 8i_2 = -4$

$i_2 = 1 + i_1 \rightarrow (2)$

2-

$v_x = -4i_1 \rightarrow (3)$

3-

$i_1 = 1.4A$, $i_2 = 2.4A$, $v_x = -4i_1 = -5.6V$

b- Find the power supplied by the dependent source

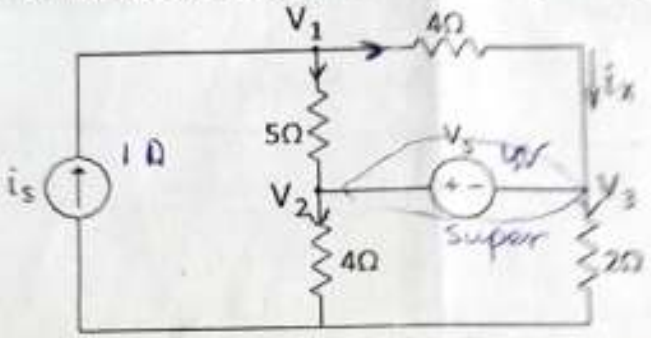
$P = \frac{v^2}{R} \Rightarrow \frac{(3 * -5.6)^2}{3} = 94.08\ w$

$P = vi = 3 * -5.6 * -2.4$
$= +40.32\ w$

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Problem 3:

Solve the circuit shown using nodal method where $i_s = 1A$, and $v_x = 4V$,



- Write all the required equations to find v_1, v_2, v_3

node 1

$$1 = \frac{v_1 - v_2}{5} + \frac{v_1 - v_3}{4} \quad \times 20$$

$$20 = 4v_1 - 4v_2 + 5v_1 - 5v_3$$

$$20 = 9v_1 - 4v_2 - 5v_3 \rightarrow (1)$$

super node

$$\frac{v_1 - v_2}{5} + \frac{v_1 - v_3}{4} + \frac{v_3}{2} + \frac{v_2}{4} = 0 \rightarrow 9v_1 - 9v_2 - 15v_3 = 0$$

$$v_2 - v_3 = 4$$

$v_1 = 4V$, $v_2 = 4V$, $v_3 = 0V$

8

8

- Find i_x

$$i_x = \frac{v_1 - v_3}{4} = \frac{4 - 0}{4} = 1A$$

i_x =

= 1A

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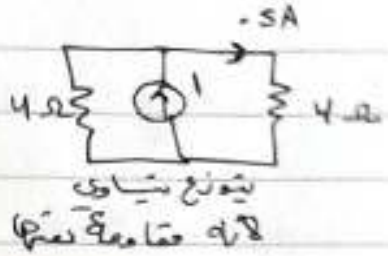
First semester 2013/2014 electric circuit

2/11/2013 First exam

problem 1 $i_s = 1A$

$(5 // 20) = 4\ \Omega$

$i_1 = \frac{-.5 * 20}{25} = -.4A$ عكس اتجاه المفروض



يتوزع تياري
لأن متساوية بمقاومتها

$v_2 = .4 * 5 = 2V$

or $i_{20} = .5 - .4 = .1\ A$

$v_{20} = .1 * 20 = 2V$

v_1 by voltage divider $= \frac{2 * 3}{4} = \frac{3}{2} = 1.5V$

problem 2 $i_s = 1A$, $v_s = 4V$

a) $-4i_1 + v_s + 3i_2 + 3v_x = 0$ but $v_s = 4V$

$-4i_1 + 3i_2 + 3v_x = -4$

2- $i_2 = i_1 + 1$

3- $v_x = -4i_1$ بحل 3 معادلات

$i_1 = 1.4A$ $i_2 = 2.4A$ $v_x = -4i_1 = -5.6V$

b) $p = vi = -2.4 * 3 * -5.6 = 40.32$ watt

problem 3 $i_s = 1A$, $v_s = 4V$

a) 1- $1 = \frac{v_1 - v_2}{5} + \frac{v_1 - v_3}{4}$

2- $v_2 - v_3 = 4$

3- $\frac{v_1 - v_2}{5} + \frac{v_1 - v_3}{4} = \frac{v_3}{2} + \frac{v_2}{4}$ | $v_1 = 4V$ $v_2 = 4V$ $v_3 = 0V$

b) $i_x = \frac{v_1 - v_3}{4v} = \frac{4.0}{4}$

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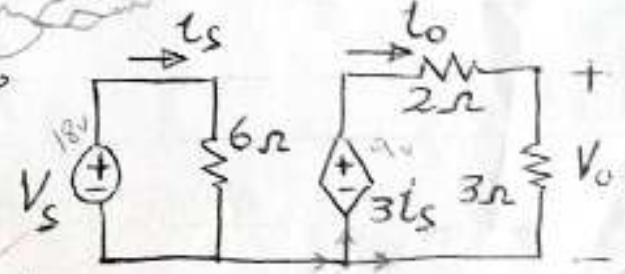
Electric Circuit I

First Exam

Summer 2012/2013

23/6/2013

1 Consider the circuit shown, where $V_s = 18V$. Find i_s, i_o, and V_o.

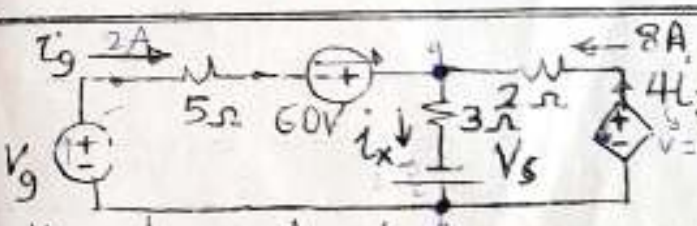


$i_s = 3A$, $i_o = 1.8A$, $V_o = 5.4V$

2 For the circuit shown, when $i_g = 2A$ find:

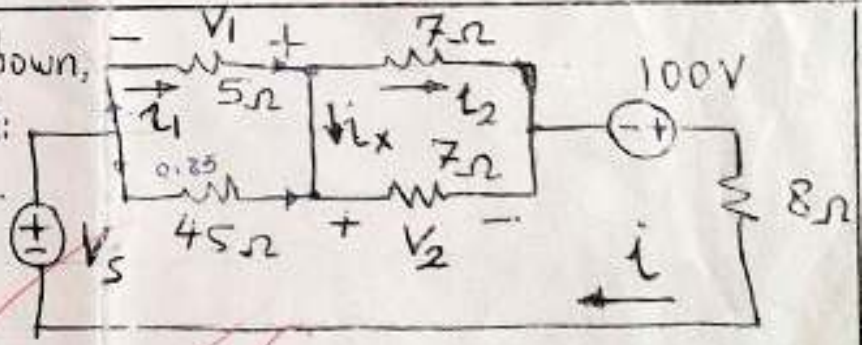
a. i_x , V_s , V_g.

b. the power supplied by the dependent source.



$i_x = 10A$, $V_s = 54V$, $V_g = -74V$, $P = 320$ watt

3 Consider the circuit shown, where $V_s = 36V$. Find: i, V_1, V_2, i_1, i_2, i_x.

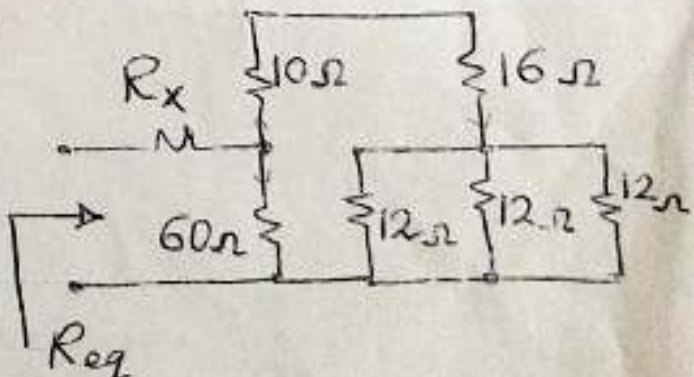


$i = 8.5A$, $V_1 = 38.25V$

$V_2 = 29.75V$, $i_1 = 7.65A$

$i_2 = 4.25A$, $i_x = 3.4A$

4 For the circuit shown, find R_{eq}, when $R_x = 20\ \Omega$.



$R_{eq} = 40\ \Omega$.

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Summer 2012/2013 electric circuit 1

23/6/2013 First exam

Q1 $i_s = \frac{v_s}{6} = 8A$ $-3i_s + 2i_o + 3i_o = 0$ $i_o = \frac{3}{5} i_s = 1.8A$

$-3i_s + 2i_o + 8v_o = 0$ $v_o = 5.4V$

Q2 $i_g + 8A = i_x$ $i_x = 10A$ because $i_g = 2A$

$-4i_x + 2x_8 + 3i_x - v_s = 0$ $v_s = 6V$

$-v_g + 5*i_g = 60 + 3*i_x - v_s = 0$

$v_g = -26V$

$P = Vi = 8 * 4 * i_x = 320 watt$

Q3 $v_s = 36V$ $(45\Omega // 5) = 4.5\Omega$ $(7 // 7) = 3.5\Omega$

$-36 + 4.5i + 3.5i - 100 + 8i = 0$ $i = \frac{136}{16} = 8.5A$

$v_{4.5} = v_1 = i * 4.5 = 8.5 * 4.5 = 38.25V$

$v_2 = v_{3.5\Omega} = 8.5 * 3.5 = 29.75V$

$i_1 = \frac{38.25}{5} = 7.65A$ $i_2 = \frac{29.75}{7} = 4.25A$

$i_x = 3.4A$

Q4 $(12 // 12 // 12) = 4\Omega$ $(4, 16\Omega) = 20\Omega$

$(10\Omega, 10\Omega) = 20\Omega$ $(30\Omega // 60\Omega) = 20\Omega$

$R_{eq} = 20 + 20 = 40\Omega$

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Al Balqa' Applied University
Faculty of Engineering Technology
Electrical Engineering Department

كلية الهندسة التكنولوجية

Student Name:

Summer Semester: 2013/2014
First Exam

Course Name: Electrical Circuits (1)
Instructors: Dr I. Abu-Harb and Eng J. Abu-Ghalune

Date: June 29, 2014
Exam Time: 45 mins.

Prob.1: Choose the correct answer.

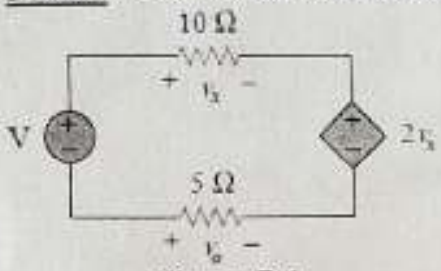


Figure B.1

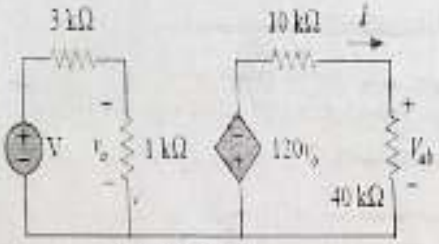


Figure B.2

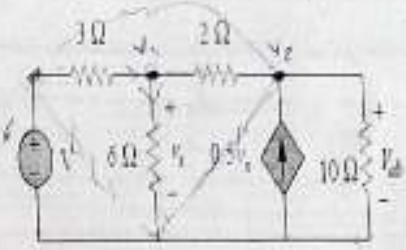


Figure B.3

***Consider the circuit shown in Fig. B.1 where* V = 7 V.**

1. The value of the current in the circuit *I* is
   a. 0.3 A b. 0.1 A **c. 0.2 A** d. 0.4 A e. None of all

2. The value of the voltage V_x is
   a. 5 V **b. 2 V** c. 3 V d. 1 V e. None of all

3. The power supplied by the dependent source is
   a. – 3.2 W b. – 1.8 W **c. – 0.4 W** d. – 0.8 W e. None of all

***Consider the circuit shown in Fig. B.2 where* V = 6 V.**

4. The value of the voltage V_a is
   a. 0.75 V b. 3 V **c. 1.5 V** d. 2.25 V e. None of all

5. The value of the voltage V_{ab} is
   a. – 54 V b. – 36 V **c. – 18 V** d. –72 V e. None of all

6. The value of the current in the circuit *I* is
   a. – 1.8 mA b. 7.2 mA **c. –0.2 mA** d. – 5.4 mA e. None of all

***Consider the circuit shown in Fig. B.3 where* V = 6 V.**

7. The value of the voltage V_x is
   a. 12 V **b. 18 V** c. 24 V d. 6 V e. None of all

8. The value of the voltage V_{ab} is
   a. 40 V b. 10 V c. 30 V **d. 20 V** e. None of all

9. The power absorbed by the 2-Ω resistor is
   a. 32 W b. 50 W **c. 8 W** d. 128 W e. None of all

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Student Name: .....

Prob. 1	1	2	3	4	5	6	7	8	9
Answer	c	b	c	c	c	c	b	d	c

8

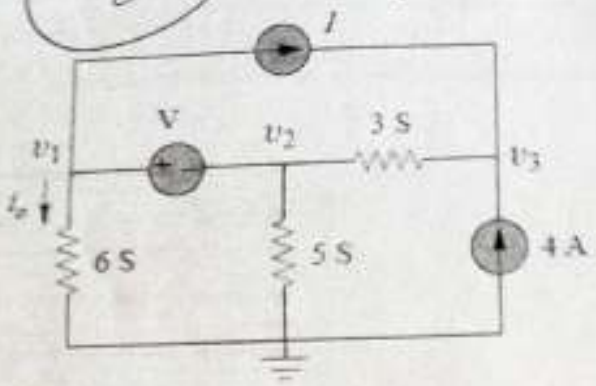


Figure B.4

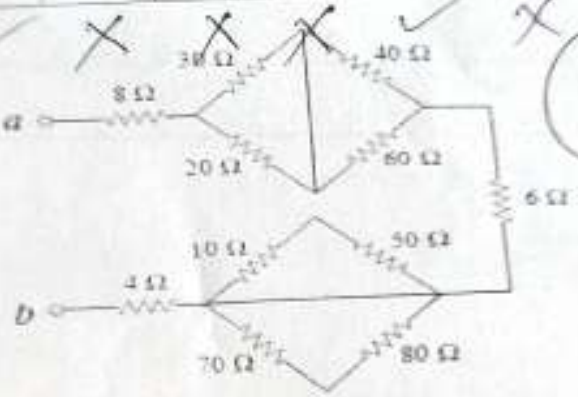


Figure B.5

Prob. 2: Consider the circuit shown in Fig. B.4 where V = 6 V and I = 1.5 A. Use nodal method in analysis.

a. Write all the required equations to calculate V_1, V_2, V_3, and i_o.

1. $i_o = 6V_1$

2. $V_1 - V_2 = 6$

3. $5V_2 + 8(V_2 - V_3) + 6V_1 = -1.5 - i_o$

4. $3(V_3 - V_2) + 5V_2 = 4$

2

b. Solve the equations to find V_1 = 2.84 V, V_2 = −3.159 V, V_3 = 3.48 V, and i_o = 17.04 A

Prob.3: Find the equivalent resistance between the *a*-*b* terminals for the circuit shown in Fig. B.5.

Solution :

$30 \parallel 20$

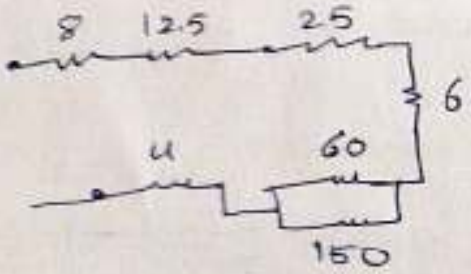
$= 12$ (5)

$40 \parallel 60$

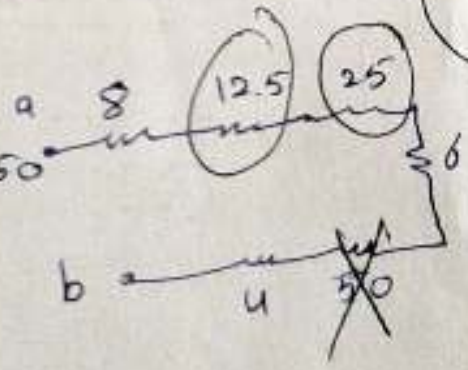
$= 24$ (25)

$60 \parallel 150$

$= 50$



2



$8 + 12.5 + 25 + 6 + 4 + 50$

$R_{eq} = 105.5\ \Omega$

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Summer semester 2013/2014 — electric circuit(1) — First exam — June 29, 2014 (B)

prob.1

1. $-7 + 10I + 2Vx + 5I = 0$ but $Vx = 10I$

so $I = \frac{7}{35} = .2A$ ©

2. $Vx = 10I = 2V$ (b)

3. $P = vi = -.2 * 2 * 2 = -.8w$ (a)

* consider the circuit show in B.2 $V = 6V$

4. $-6 + 4I = 0$ $I = \frac{6}{4} = 1.5mA$ $v_o = 1 * 1.5 = 1.5V$ ©

5. $120v_o + 10I + 40I = 0$ but $v_o = 1.5$ $I = -\frac{180}{50} = -3.6mA$

$V_{ab} = -3.6 * 40 = -144V$ (e)

6. $I = -3.6mA$ (e)

* Consider the circuit shown in Fig B.3 $V = 6V$

7. by using nodal $\frac{6 - Vx}{3} = \frac{Vx}{6} + \frac{Vx - Vab}{2}$

and $\frac{Vx - Vab}{2} + .5Vx = \frac{Vab}{10}$ by solving eq

$Vx = 12V$ (a)

8. $V_{ab} = 24V$ (d)

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9. $P = \frac{V^2}{R} = \frac{(12-20)^2}{2} = 32$ watt ②

prob 2 $V = 6V$ $I = 1.5A$ use nodal method

a. 1. $6v_1 + 5v_2 + I + 3(v_2 - v_3) = 0$ but $I = 1.5A$

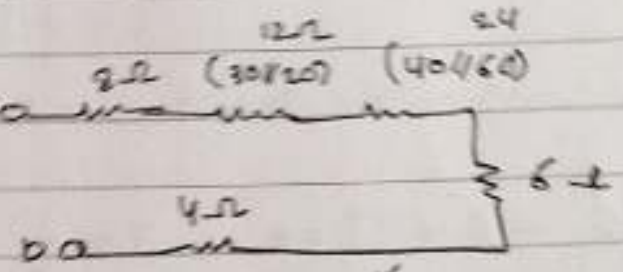
$6v_1 + 5v_2 = -1.5A - (v_2 - v_3) * 3$

2. $v_1 - v_2 = 6$

3. $4 + 1.5 + 3(v_2 - v_3) = 0$

b $v_1 = 3.09V$ $v_2 = -2.91V$ $v_3 = -1.075V$ $i_o = 18.9mA$

Prob 3 Req



Req = 54 Ω

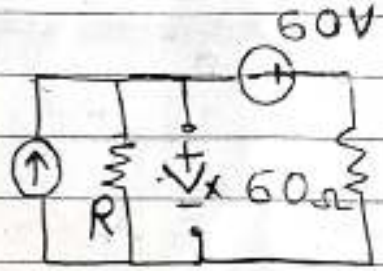
اختاروا حاجة تخلي
الدائرة سهلة بتخلي
الحسابات فارغ

Electric Circuit I

Second exam.
14/7/2013

Write all the required equations that are used to solve the following problems.

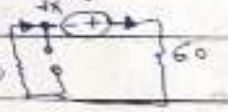
1 Using superposition principle, find V_x when $I_s = 8\,A$, $R = 30\,\Omega$.



a. V_x due to voltage source only.

Eq. $\frac{-V_x}{30} = \frac{V_x + 60}{60} \rightarrow -3V_x = 60$

$V_x = -20$ volt



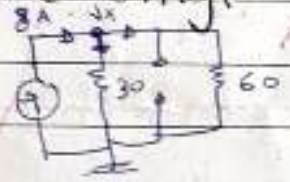
$V_x = -20$ volt

b. V_x due to current source only.

Eq. $8 = \frac{V_x}{30} + \frac{V_x}{60}$

$3V_x = 480$

$V_x = 160$ volt



$V_x = 160$ volt

c. V_x due to both sources.

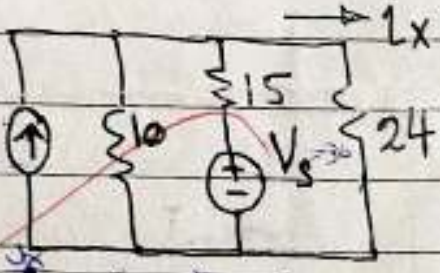
Eq. $V_{x\,total} = V_x + V_x$

$-20 + 160 = 120$ Volt

$V_x = 120$ volt

2 Using source transformation method, find i_x.

$V_s = 30V$, $I_s = 8\,A$.

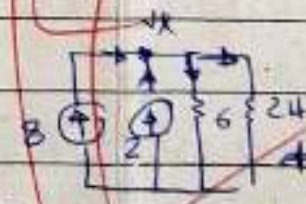


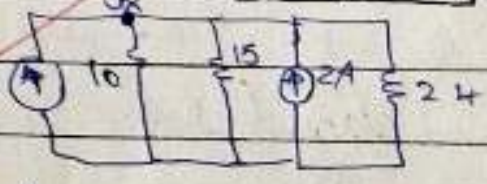
Circuit

$8 + 2 = \frac{V_x}{6} + \frac{V_x}{24}$

$10 = \frac{5V_x}{24}$

$V_x = 48$





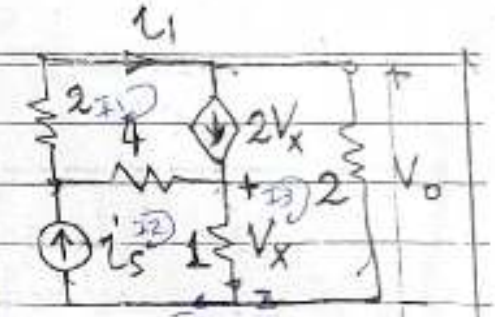
$i_x = \frac{V_x}{24} = \frac{48}{24} = 2A$

Eq. $i_x = 2A$

$i_x = 2A$

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3 Using mesh method, find i_1 and V_o when $i_s = 8$ A.



$(i_x = I_2 - I_3)$ $V_x = i_x(1)$

1) $I_1 - I_3 = 2V_x$ → $I_1 + I_3 - 2I_2 = 0$

Super mesh Loop

2) $2I_1 + 2I_3 + 1(I_3 - I_2) + 4(I_1 - I_2) = 0$ → $6I_1 - 5I_2 + 3I_3 = 0$

3) $I_2 = 8$ A

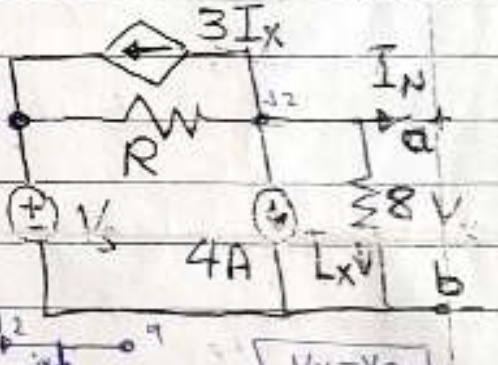
$i_1 = -2.66$ A

V_o → 4) $6I_1 + 3I_3 = 40$, $I_1 + I_3 = 16$ → $I_1 = 2.66$ A, $I_3 = 18.6$ A

$V_o = 37.2$ volt

$V_o = 2I_3 \rightarrow 2(18.6) = 37.32$

4 Consider the circuit shown where $R = 4\ \Omega$, $V_s = 20$ V.



a. Using nodal method find V_x.

at node 1 $V_1 = 20$ volt $\left\{ i_x = \frac{-V_2}{8} \right.$

at node 2 $I_2 = 4 + 3\left(\frac{V_2}{8}\right) + \frac{V_2}{8}$

$\frac{V_1 - V_2}{4} = 4 + \frac{4V_2}{8}$

$\frac{20 - V_2}{4} = \frac{4V_2}{8} + 4$

$5 - \frac{V_2}{4} = \frac{4V_2}{8} + 4$ → $5 - 4 = \frac{4}{8}V_2 + \frac{V_2}{4} \rightarrow 1 = \frac{3V_2}{8} \rightarrow V_2 = \frac{8}{3} = 2.6$

$V_x = V_2$

$V_x = 2.66$ volt

b. Short circuit the a-b terminals ($V_x = 0$) then using mesh method find I_N.

$I_1 - I_2 = 4$

$-20 + 4(I_1 - I_4) + 8(I_2 - I_3) = 0$

$8(I_3 - I_2) = 0$

$I_4 = 3(I_2 - I_3) = 0$

$I_4 - 3I_2 + 3I_3 = 0$

$I_4 = 3I_x$

$I_x + I_3 = I_2$

$I_x = I_2 - I_3$

$I_3 = I_N$

$I_N = 1$ A

c. Find the Thevenin's equivalent resistance R_{th} between the a-b terminals.

$I_1 - 3I_2 + 3I_3 = 0$

$4I_1 + 4I_2 - 8I_3 = 0$

$-3I_2 + 8I_3 = -1$

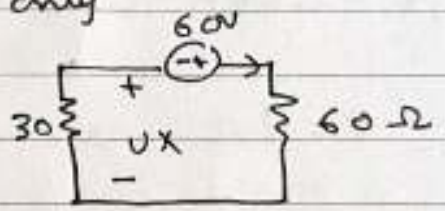
$R_{th} = \frac{1}{0.235}$ → $R_{th} = 4.255\ \Omega$

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Summer 2012/2013 electric circuit 1
14/7/2013 Second exam

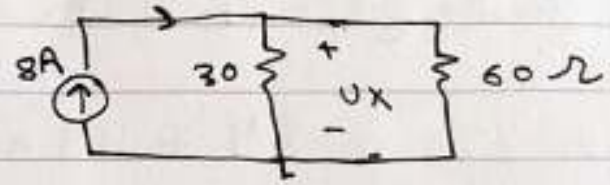
Q1 $I_s = 8A$, $R = 30\Omega$ Find v_x

a. v_x due the voltage source only



$$V_x = \frac{-60 * 30}{90} = -20V$$

b. v_x due the current source only



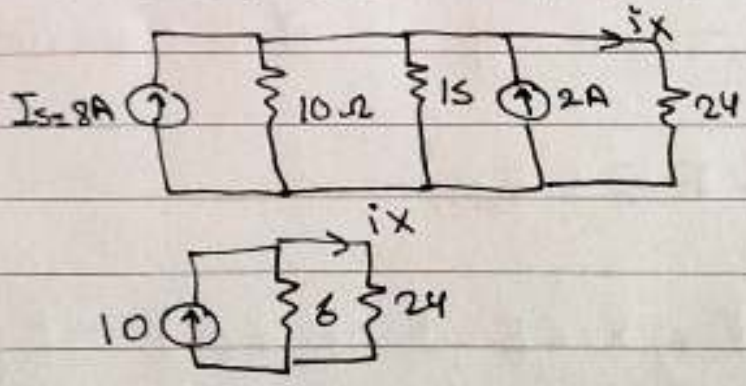
$$V_x = \frac{8 * 60}{90} * 30 = 160V$$

c. v_x due to both sources

$$V_x = -20 + 160 = 140V$$

Q2 Find i_x $v_s = 30V$ $I_s = 8A$

حول v_s و i_s الى مصدر تيار موصولة على التوازي



$$i_x = \frac{10 * 6}{30} = 2A$$

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Q3 Find i_1 and v_o when $i_s = 8A$

1) $6i_1 + 3i_3 - 5i_2 = 0$
2) $i_2 = 8A$
3) $i_1 = 2v_x + i_3 \Rightarrow i_1 = 2i_2 - i_3$
4) $v_x = 1(i_2 - i_3)$

$i_1 = -2.66A$ $i_3 = 18.66A$ $v_o = 2i_3 = 37.33V$ وبحل المعادلات

Q4 $R = 4\Omega$, $V_s = 20V$ Find v_x by nodal

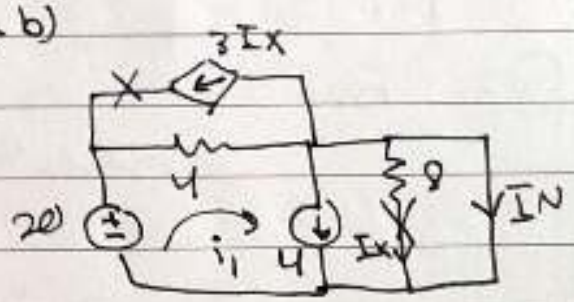
$V_1 = V_s = 20V$

node 2 $\quad 4 + 3I_x + I_x = \frac{20 - v_x}{4}$ but $V_x = 8I_x$

$4 + 4I_x = 5 - 2I_x$

$6I_x = 1 \quad I_x = .166A \quad V_x = 1.33$

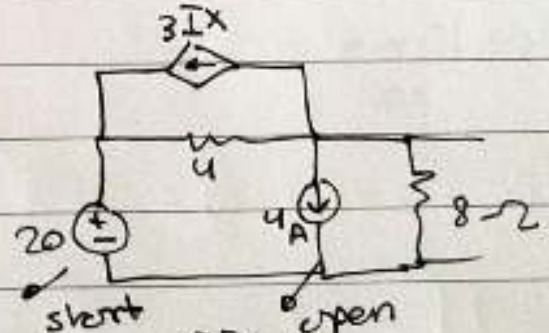
b. Find I_N between (a,b)



$I_N = \frac{20}{4} = 5A$

because $I_x = 0$

c. Find R_{th}



$3I_x + I_x + I = 1$

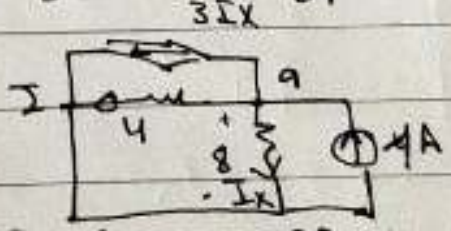
$4I_x + I = 1$

$-4I + 8I_x = 0$

$I = 2I_x$

$I_x = .5IA$

$I_x = \frac{1}{6} \quad v = 8 * \frac{1}{6} = 1.33 \quad R_{th} = 1.33\Omega$



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Al Balqa' Applied University
Faculty of Engineering Technology
Electrical Engineering Department

كلية الهندسة التكنولوجية

Student Name: .....

First Semester: 2013/2014
First Exam

Course Name: Electrical Circuits (1) **Date:** Dec. 14, 2014
Instructors: Dr. M. Dwairi, Dr. I. Abu Harb, Eng. J. Abu-Galune **Exam Time:** 50 minutes

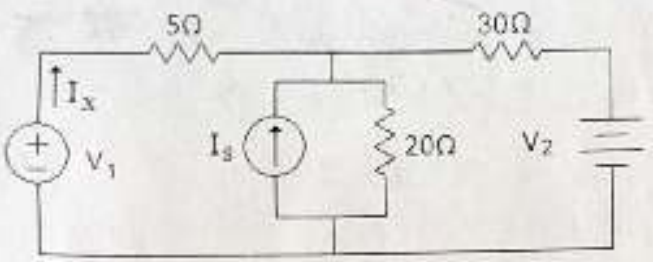


Figure A.1

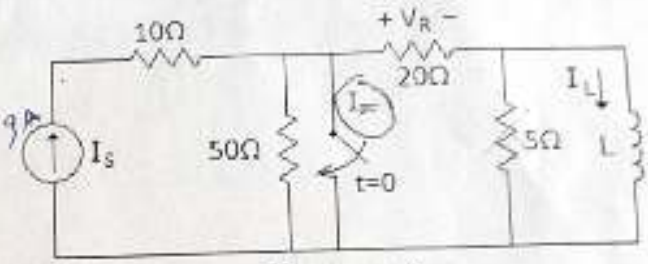


Figure A.2

Problem 1: Using superposition method, find $\mathbf{I_x}$ for the circuit shown in Fig. A. 1where I_s = 3 A, V_1 = 6V, and V_2 = 8V according the following steps:

a. Due to the left voltage source $\mathbf{V_1}$:

$\mathbf{I_{x_1}}$ = $\frac{V}{R_{eq}}$ = $\frac{6}{17}$ = 0.35 A

b. Due to the current source $\mathbf{I_s}$:

$\mathbf{I_{x_2}}$ = $-I_s \times \frac{R_{2\Omega}}{R_{2\Omega}}$ = $-3 \times \frac{12}{17}$ = −2.11 A

c. Due to the right voltage source $\mathbf{V_2}$:

$\mathbf{I_{x_3}}$ = $\frac{V}{R_{eq}}$ = $\frac{8}{34}$ = 0.18 = I_x

d. Due to the all sources:

$\mathbf{I_x}$ = $I_{x_1} + I_{x_2} + I_{x_3}$ = 0.35 + −2.11 + −0.18 = −1.94 A

Problem 2: Consider the circuit shown in Fig. A.2 where the switch is opened for a long time and closed at t = 0. Assume L = 0. 5 H and I_s = 3A, then complete:

a. For t < 0:

I_L = $\frac{3 \times 5\Omega}{7\Omega}$ = 2.14 A

b. For t > 0:

1. The time constant (τ) = $\frac{L}{R_{eq}}$ = $\frac{0.5}{4}$ = $\frac{1}{8}$ s
2. I_L = $I_{L(0)} e^{-t/\tau}$ = $2.14 e^{-8t}$
3. I_{sw} = ...

$V_L = V_R = L \frac{di}{dt} = 0.5 \times 2.14 \times -8 e^{-8t} = -8.56 e^{-8t}$ V

$I_{sw} = \frac{V_R}{20} = \frac{-8.56 e^{-8t}}{20} = -0.428 e^{-8t}$ A

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Problem 3: Consider the circuit shown in Fig. A.3 where the switch is opened for a long time and closed at t = 0. Assume C = 5 F and I_s = 3 A, then complete :

a. For t < 0:

1. V_R = $i \times 20$ = 40 V

2. The energy stored in the capacitor W_C = $\frac{1}{2} C V_0^2$ = $\frac{1}{2} \times 5 \times (40)^2$ = 4000 J

b. For t > 0:

1. The time constant (τ) = RC = 4×5 = 20 s

2. $V_R = V_c = V_0 e^{-t/\tau}$ = $40 e^{-0.05t}$ V

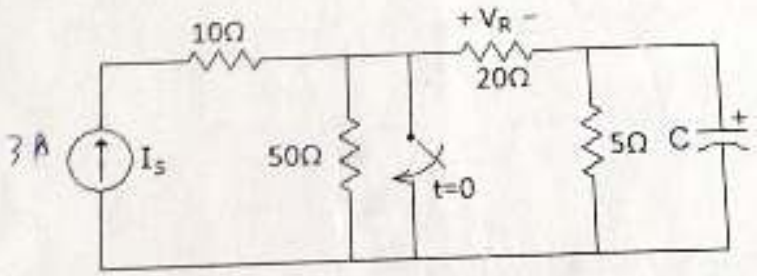


Figure A.3

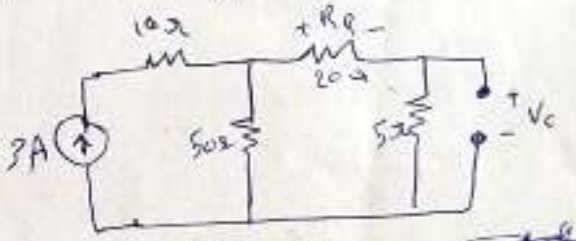
b) t > 0

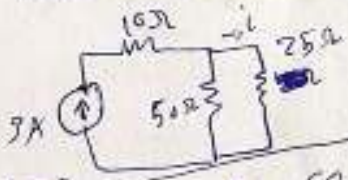


$R_{eq} = \frac{20 \times 5}{25} = 4\Omega$

$V_R = V_c = V_0 e^{-t/\tau} = 40 e^{-\frac{1}{20} t}$

t < 0





a) $i = 3 \times \frac{50}{75}$ = 2 A

$V_R = i \times 20$ =

$= 2 \times 20 = 40$ V

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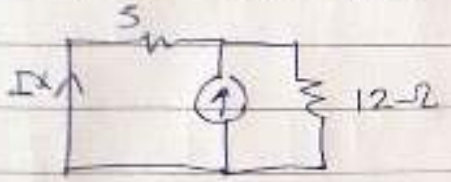
First semester 2013/2014 — Electrical circuit I second exam — 14/Dec/2014 — A

problem 1

a. Due to the left voltage source V1

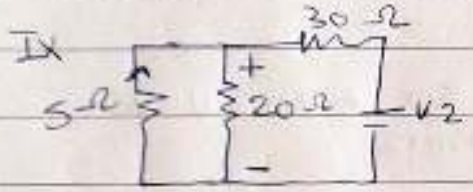
$(30 // 20) = 12\ \Omega$ $\quad I_x = \frac{6}{12+5} = 0.353A$

b. Due to the current source Is



$I_x = \frac{-3 * 12}{12+5} = -2.1176\ A$ (عكس اتجاه)

C. Due to the right voltage source V2



$V_{20\Omega} = V_{5\Omega} = \frac{8 * 4}{34} = .941\ V$

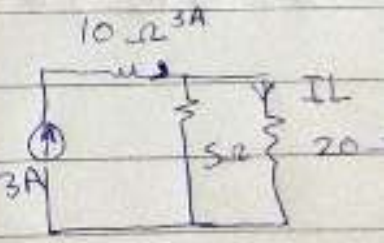
$I_x = -\frac{.941}{5} = -.188A$ (الإشارة لسالبة عشان عكس اتجاه الي رسمه وهو يمين عكس عقارب)

d. Due to the all sources

$I_x = I_{x1} + I_{x2} + I_{x3} = 0.35 - 2.1176 - .188 = -1.95$

problem 2

a. For t<0 $\quad I_L = \frac{3 * 50}{70} = 2.14A$

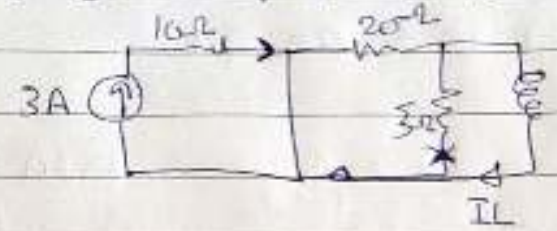


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b. For $t > 0$

1. The Time constant $(\tau) = \frac{L}{R_{eq}} = \frac{.5}{20//5} = .125\,S$

3. I_{sw} لإيجاد قيمة I_{sw} نستبدل المحث بمصدر تيار

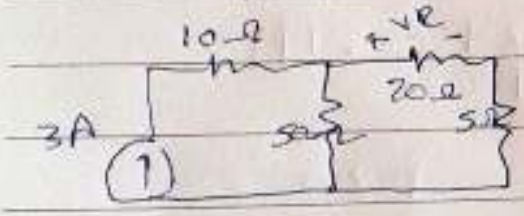


$I_{sw} = 3 - I_L = 3 - 2.14 = .86\,A$

2. $I_L = I_{L(0)}\, e^{-\frac{t}{\tau}} = 2.14\, e^{-8t}$

problem 3

a. For $t < 0$



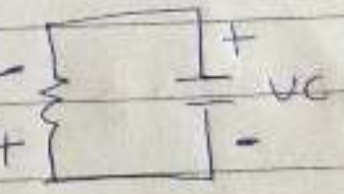
$V_R = i_R * R = \frac{3 * 50}{50+20+5} * 20 = 40V$

2. The energy stored in capacitor $e = \frac{1}{2} C V_0^2 = \frac{1}{2} * 5 * (10)^2 = 25$

b. For $t > 0$

1 The Time constant $(\tau) = RC = (20//5) * 5 = 20\,S$

2. $V_R = -V_0\, e^{-\frac{t}{\tau}} = -10\, e^{-\frac{t}{20}}$



(∵ V_C ...

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Final exam 4-08-2015

Q1: Find the voltage (v) for the network shown in Fig 1 using superposition theorem. (10marks)

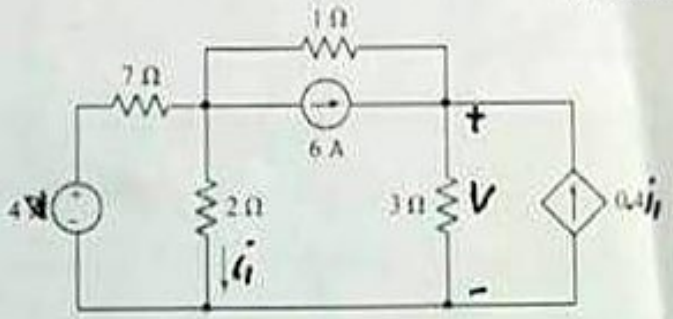


Fig 1

Q2: For the circuit shown in Fig 2: (10marks)

a) Determine the value of Rl to which a maximum power can be delivered .

b) The maximum power deliverd to Rl.

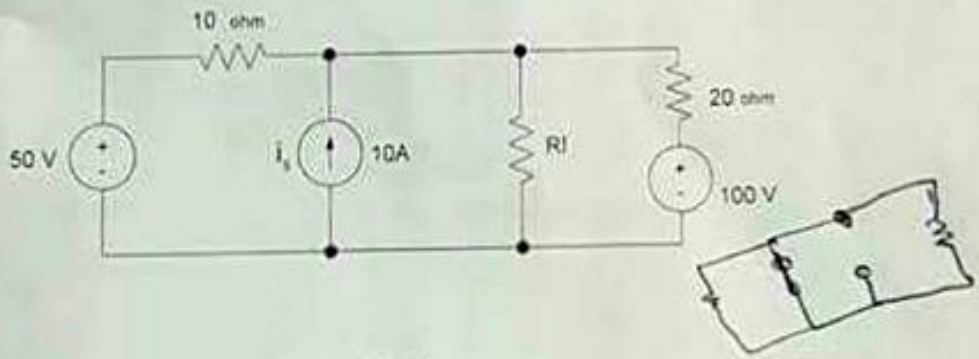


Fig 2

Q3: a) For the circuit shown in Fig 3.a find the equivalent capacitance (10 marks)

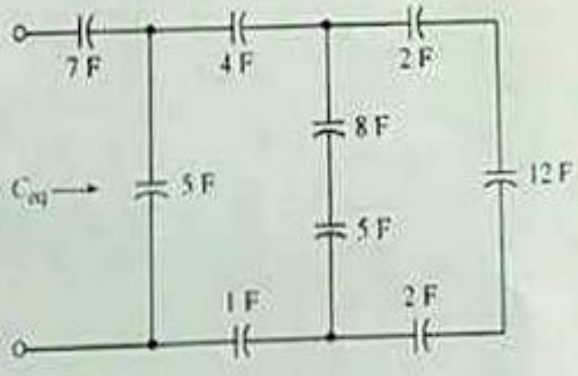


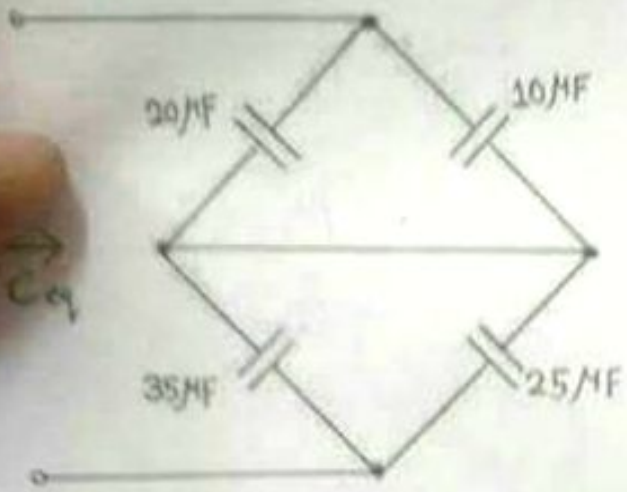
Fig 3.a

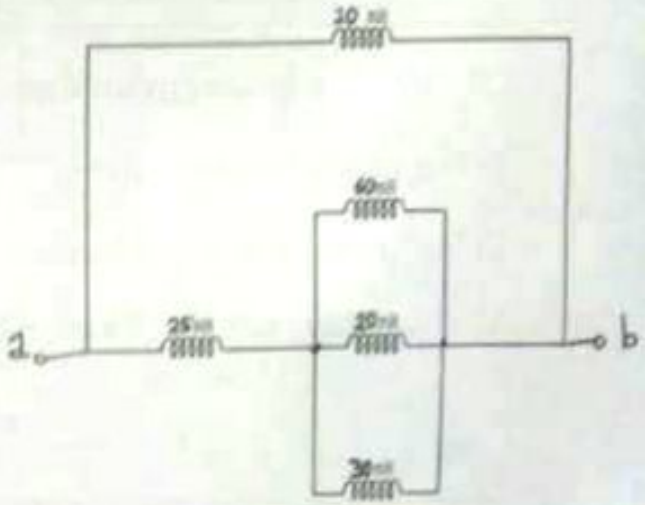
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1) For the previous circuit, Determine the Thevenin and Norton equivalents for the given network.

2) Find the resistance that dissipate maximum power if it's connected between a and b terminals.

Q4: For the given circuits, calculate Ceq and Leq .





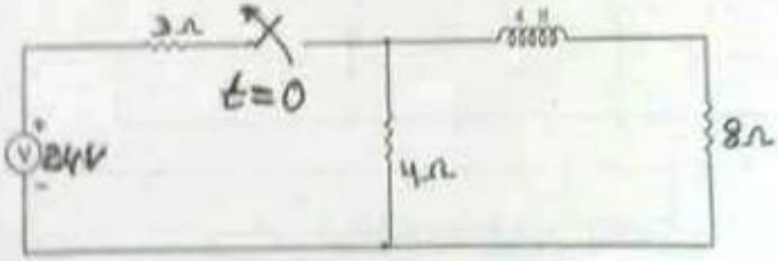
Q5: For the given circuit below find the current in the inductor at:

1) t= 0⁻ (t < zero).

2) t= ∞

3) t= 2s

4) The energy dissipated in 8Ω resistor.



End of examinations papers.

Dr.Yousif Eltous --- Dr. Ibrahim Abu Harb

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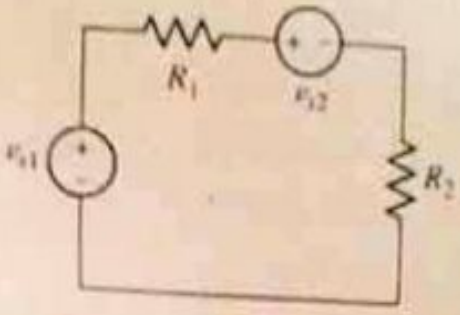
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كلية الهندسة التكنولوجية

First exam 25-06-2015

Problem 1.[9 points]

In the circuit of figure, V_{s1} =100V, V_{s2}=40V, R_1=20Ω, and R_2=10Ω.Compute:

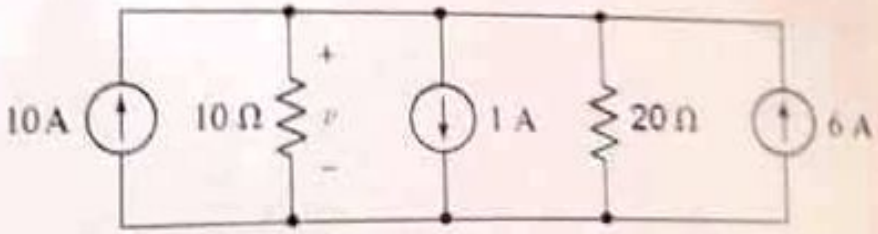
1- The voltage across R_1 (V_{R1}).
2- The current through the circuit.
3- The power absorbed by V_{s1} and V_{s2}.
4- The power absorbed by R_2.



Problem 2.[7 points]

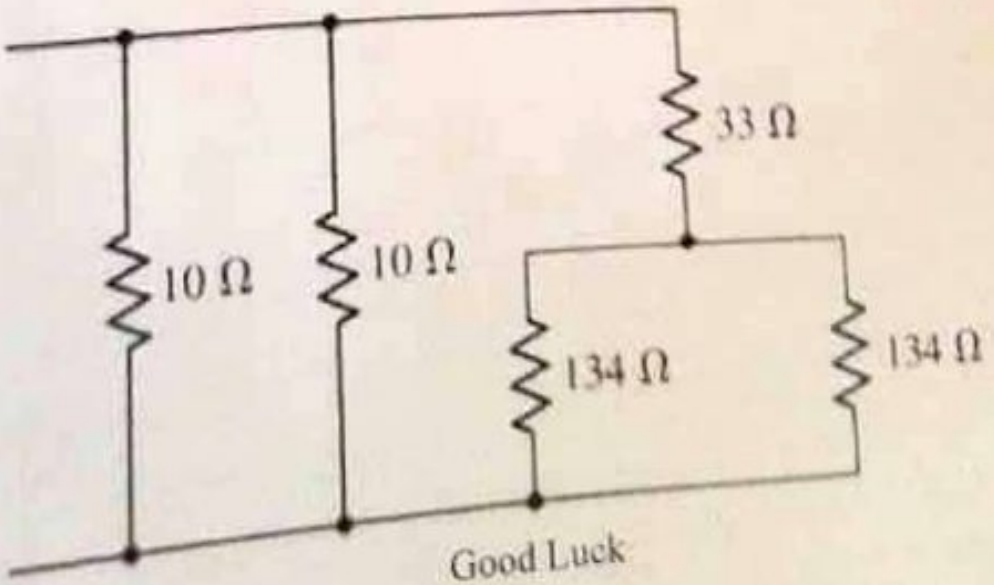
For the single node pair circuit shown in figure determine:

1- The voltage V.
2- The current through 10 Ω resistor
3- The power supplied by 10A current source



Problem 3.[4 points]

Find the equivalent resistance for the network shown in figure



Good Luck

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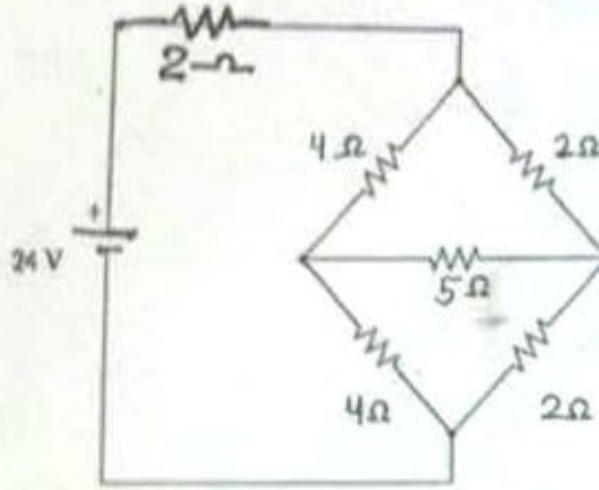
Electric Circuits – *final* Exam – May 20th , 2015

Q1: For the given circuit find:

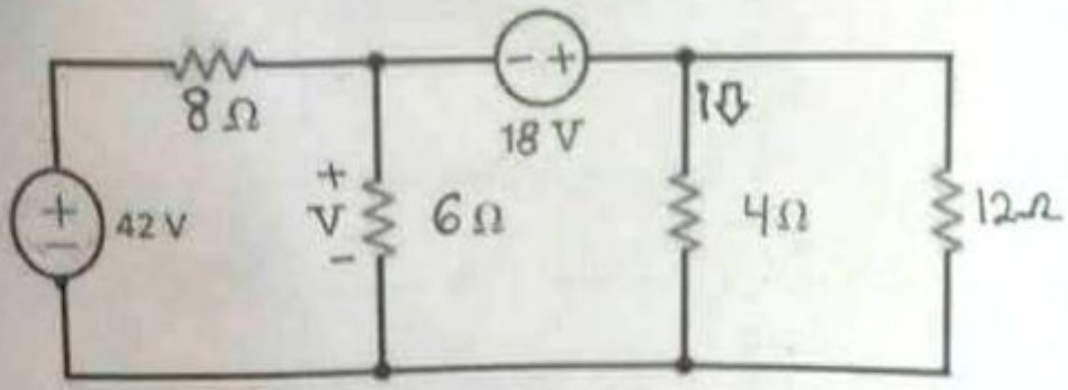
1) The equivalence resistance.

2) The power generated by the source.

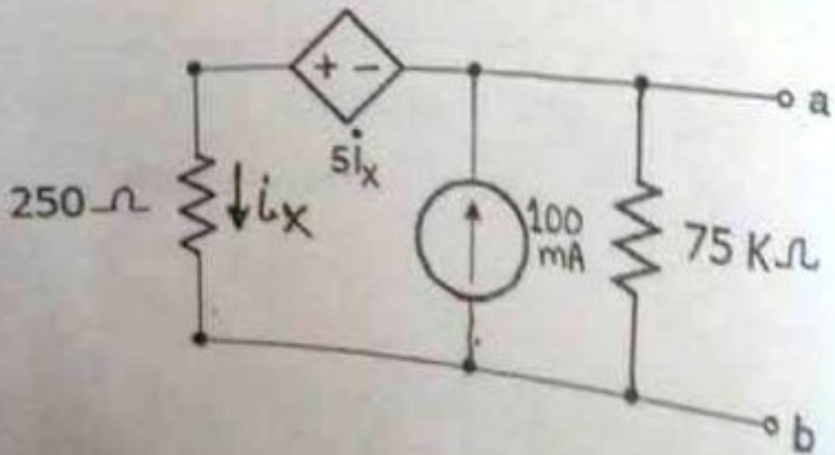
3) The voltage across 5Ω resistor.



Q2: Using Nodal analysis, find V and i .



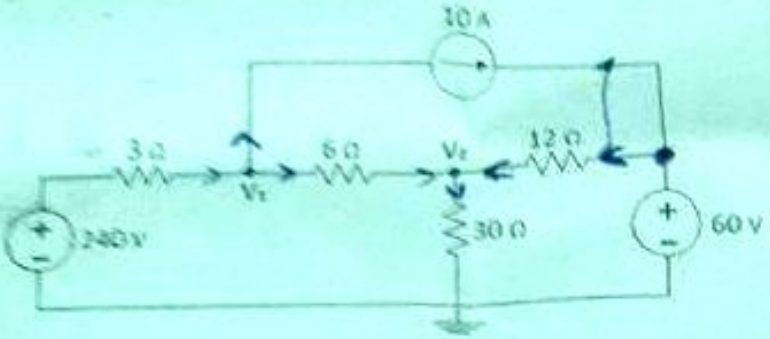
Q3: Answer the questions next page.



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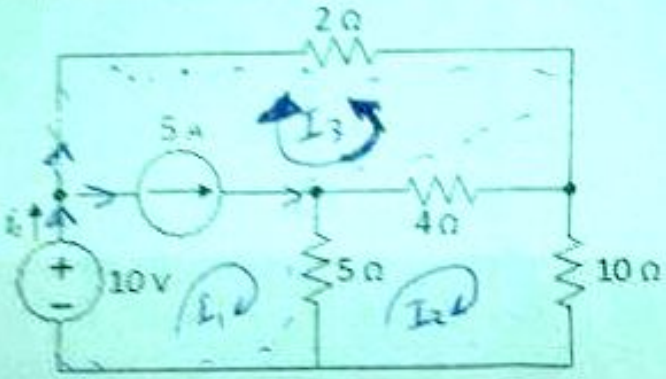
Question One:

Use nodal analysis to determine V_1 and V_2.



Question Two:

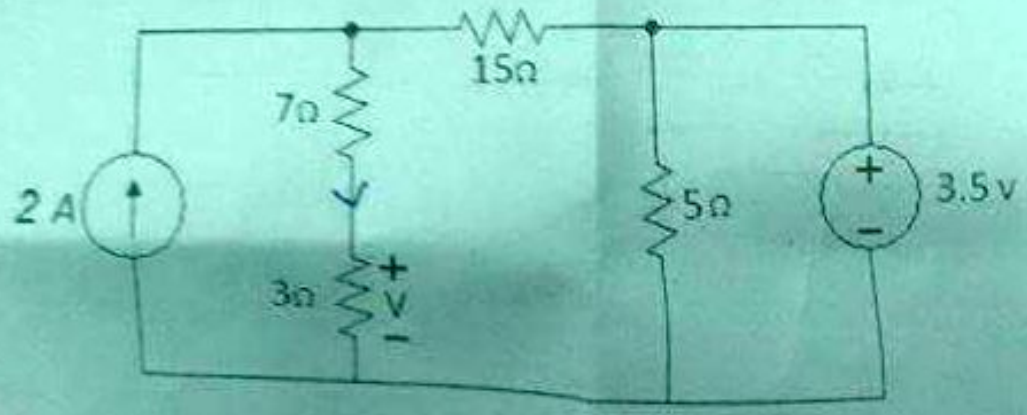
Use mesh analysis to calculate the current of the voltage source i_s.



Question Three:

Use superposition principle to compute the voltage V and the power absorbed by th

5 - Ω resistor.



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b) Calculate the power dissipated in the 40 Ω resistor and the voltage labeled Vc for the circuit given in fig 3.b

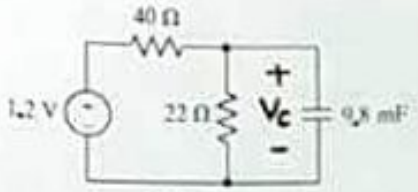
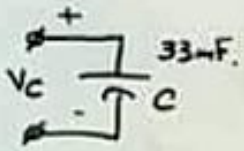


fig 3.b

Q4: The voltage across a 33 mF capacitor is shown graphically in Fig 4 : (8 marks)

a) Assuming the passive sign convention, sketch the resulting current waveform across the capacitor.

b) Find the energy stored in the capacitor at t=20ms.



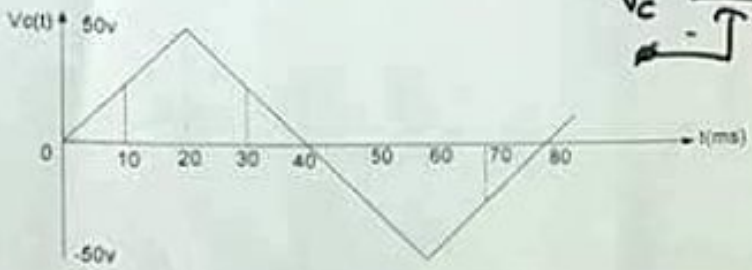


Fig 4

Q5: For the circuit shown in Fig 5 the switch opens at t=0 after having been closed for a long time. Find: (12 marks)

a) $i_1(t)$ for $t >= 0$.
b) $i_L(t)$ for $t >= 0^+$.
c) $V_2(t)$ for $t >= 0^+$.

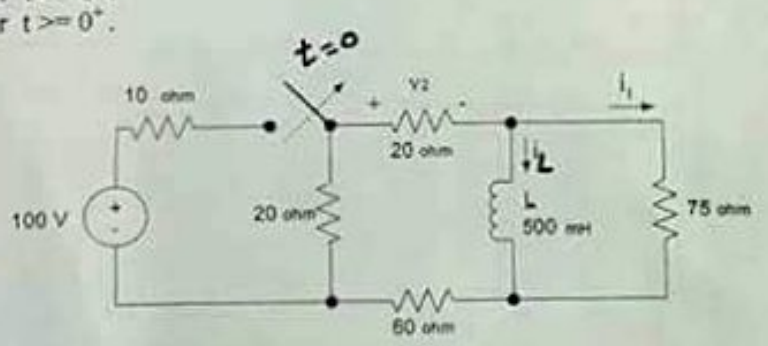


Fig 5

Good Luck

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